CSCI 361: Exercise # 12

Your Name: ____________________

Due: Thursday, Oct 30, in class

Assigned reading for this lecture:

- Sipser Third Edition Chap 5.2 (We will skip 5.1)

(Reduction via Computation Histories and PCP)
To prove that some properties of the language of CFGs are undecidable, we need a new reduction that encodes the computation histories of TMs into a puzzle about strings. Read the definition of the Post Correspondence Problem and answer the following.

Questions: Consider a small instance of PCP with three types of dominos (listed in order)

$$\left\{ \frac{a}{ab}, \frac{abb}{b}, \frac{b}{a} \right\}$$

1. Does this instance have a valid solution? What is the **sequence of indices** such that concatenating the top and bottom strings of the corresponding dominoes gives a match?

2. Read the Proof Idea of the reduction. To reduce valid TM computations to the PCP problem, conceptually, what should a match in the dominoes force?